

1 Turing Machines

Definition A Turing Machine is a finite state machine with an infinite and unrestricted memory (denoted as “tape”).

Differences from finite automata:

- A TM can both read and write on the tape.
- The tape head can move both left and right.
- The tape is infinite.
- The accept/reject states take immediate effect.

Note: A TM may not stop on every input.

Church-Turing Thesis: Everything “computable” is computable by a TM.

One transition of a TM:

- Change the state of M .
- Write a symbol on the tape.
- Move the head one cell to the left or right.

Definition A TM is an 8-tuple $(Q, \Sigma, \Gamma, B, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$ where:

- Q is a finite set of states.
- Σ is the input alphabet.
- $\Gamma \supseteq \Sigma$ is the tape alphabet.
- $B \in \Gamma - \Sigma$ is a special *blank symbol* (also denoted by $\sqcup$).
- $\delta : Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$ is the transition function.
- $q_0 \in Q$ is the start state.
- $q_{\text{accept}} \in Q$ is the accepting state.
- $q_{\text{reject}} \in Q$ is the rejecting state.

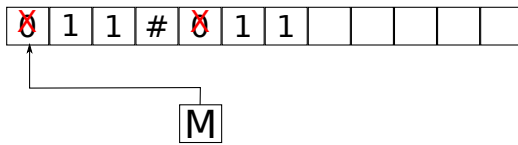
Definition If the TM is in state q , and the tape head is over symbol a , then the transition:

$$\delta(q, a) = (r, b, L)$$

tells the TM to move to state r , replace a with b , and move the head left.

Example A TM that accepts

$$L = \{w\#w \mid w \in \{0,1\}^*\}$$



M_1 = on input string w

1. Zig-zag across the tape to corresponding positions on either side of the $\#$ symbol to check whether these positions contain the same symbol. If they do not, or if no $\#$ is found, *reject*. Cross off symbols as they are checked to keep track of which symbols correspond.
2. When all symbols to the left of the $\#$ have been crossed off, check for any remaining symbols to the right of the $\#$. If any symbols remain, *reject*; otherwise, *accept*.

Consider $w = 011000\#011000$

$\rightarrow 0$	1	1	0	0	0	#	0	1	1	0	0	0	$\sqcup$	...
x	$\rightarrow 1$	1	0	0	0	#	0	1	1	0	0	0	$\sqcup$	...
x	1	1	0	0	0	#	$\rightarrow x$	1	1	0	0	0	$\sqcup$	...
$\rightarrow x$	1	1	0	0	0	#	x	1	1	0	0	0	$\sqcup$	...
x	$\rightarrow x$	1	0	0	0	#	x	1	1	0	0	0	$\sqcup$	...
x	x	x	x	x	x	#	x	x	x	x	x	x	$\rightarrow \sqcup$	...
													accept	...

Configuration of a TM is specified by three things:

- state of the machine.
- contents of the tape.
- position of the tape head.

It is written as “ w_1qw_2 ”, where $w_1, w_2 \in \Gamma^*$ and $q \in Q$, to denote:

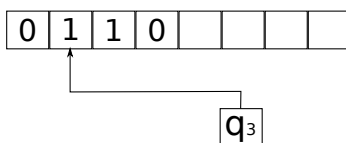
- the machine is in state q
- the tape has string w_1w_2
- the tape head is on the leftmost symbol of w_2

Definition Configuration C_1 **yields** C_2 if the TM can go from C_1 to C_2 .

- $C_1 = uaq_i bv$ yields $C_2 = uq_j acv$ if $\delta(q_i, b) = (q_j, c, L)$
- $C_1 = uaq_i bv$ yields $C_2 = uacq_j v$ if $\delta(q_i, b) = (q_j, c, R)$

Definition The start configuration of a TM for an input string w is denoted as q_0w .

Example



is denoted as $0q_3110$

1.1 Turing-recognizable and decidable languages

Three possible outcomes for an input w :

- accept
- reject
- infinite loop

Accept and *Reject* states are *halting* states.

Definition A TM *accepts* w through configurations $C_1 C_2 \dots C_k$ if:

- C_1 is the starting configuration.
- C_i yields C_{i+1} , $1 \leq i \leq k - 1$
- C_k is the accepting configuration.

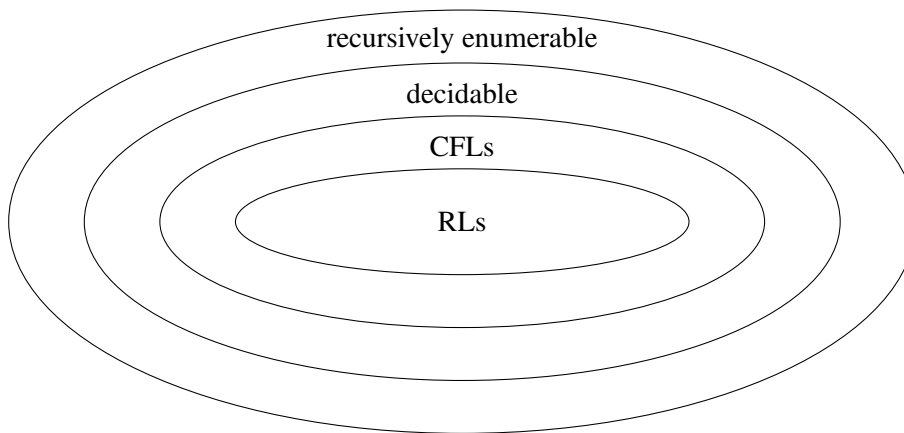
Definition The language of a TM M , denoted by $L(M)$, is the set of all strings accepted by M .

Definition Languages that are recognized/accepted by some TM are called *Turing-recognizable*, or *recursively enumerable* languages.

Definition A TM that halts on every input is called a *decider*.

Definition The languages that are accepted by decider TMs are called *Turing-decidable*, or *decidable*, or *recursive* languages.

Definition



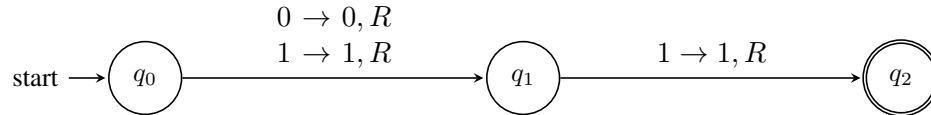
Example

- $L(M_1) = \{w \mid \text{second character of } w \text{ is } 1\}$

Informal description:

1. Read the first letter, move to the right.
2. Read the second letter. If it is a 1, accept. Else, reject.

Formal description (transition diagram):

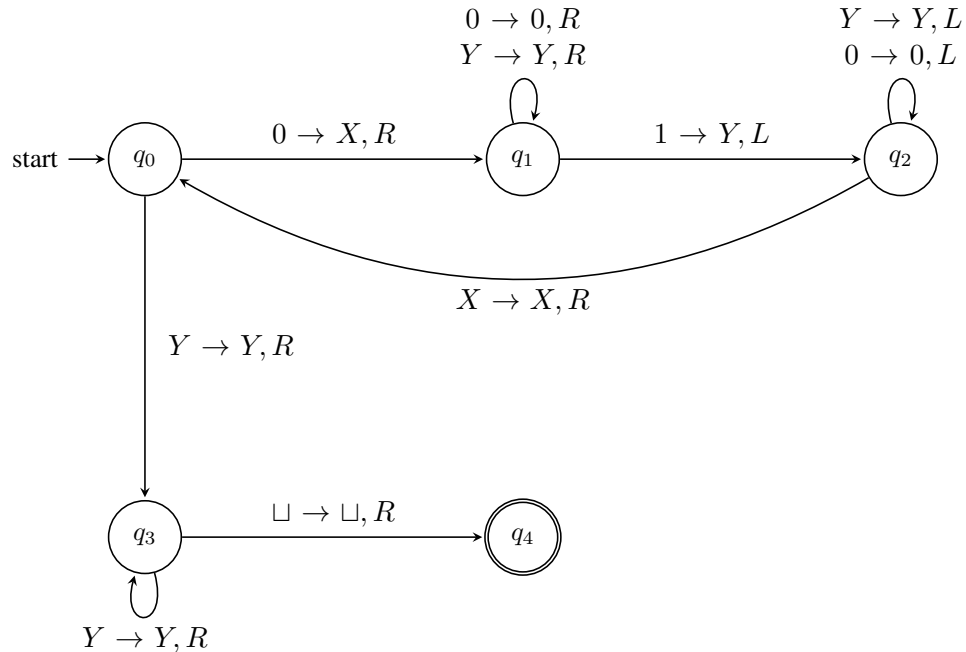


- $L(M_2) = \{0^n 1^n \mid n \geq 1\}$

Informal description:

1. Check the leftmost 0, replace it by X .
2. Move to the right, skip 0s and Y s. If a 1 is found, replace it with Y .
3. Move to the left until you find an X . If an X is found, move one cell to the right.
4. If that symbol is 0, repeat the above procedure. If that symbol is Y , move to the right, to the end of Y s. If you find a $\sqcup$, accept.

Formal description (transition diagram):



- $L(M_3) = \{a^i b^j c^k \mid k = i \times j\}$

Informal description:

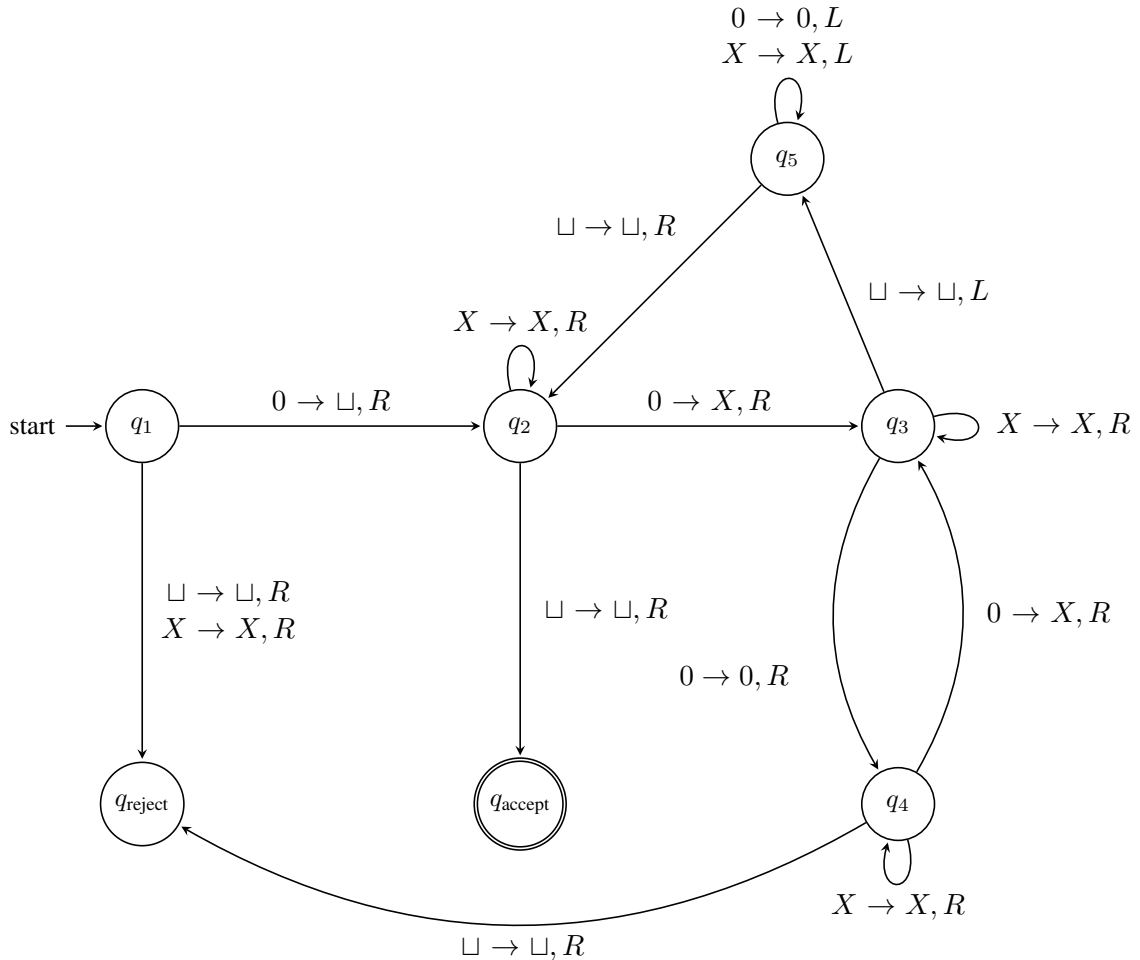
1. Find the leftmost a , replace it with X .
2. For every b :
 - (a) replace the b with Y .
 - (b) replace one c with Z .
3. Move to the left to the leftmost a , replace each Y with a b .
4. Repeat the above procedure for each remaining a , replace a with X .
5. When no a can be found, check the end of the string. If no c is left, accept.

- $L(M_4) = \{0^{2^n} \mid n \geq 0\}$

Informal description:

1. Sweep left to right across the tape, crossing off every other 0.
2. If in stage 1 the tape contained a single 0, *accept*.
3. If in stage 1 the tape contained more than a single 0 and the number of 0s was odd, *reject*.
4. Return the head to the left-hand end of the tape.
5. Go to stage 1.

Formal description (transition diagram):



Sample Run: $w = 0000$

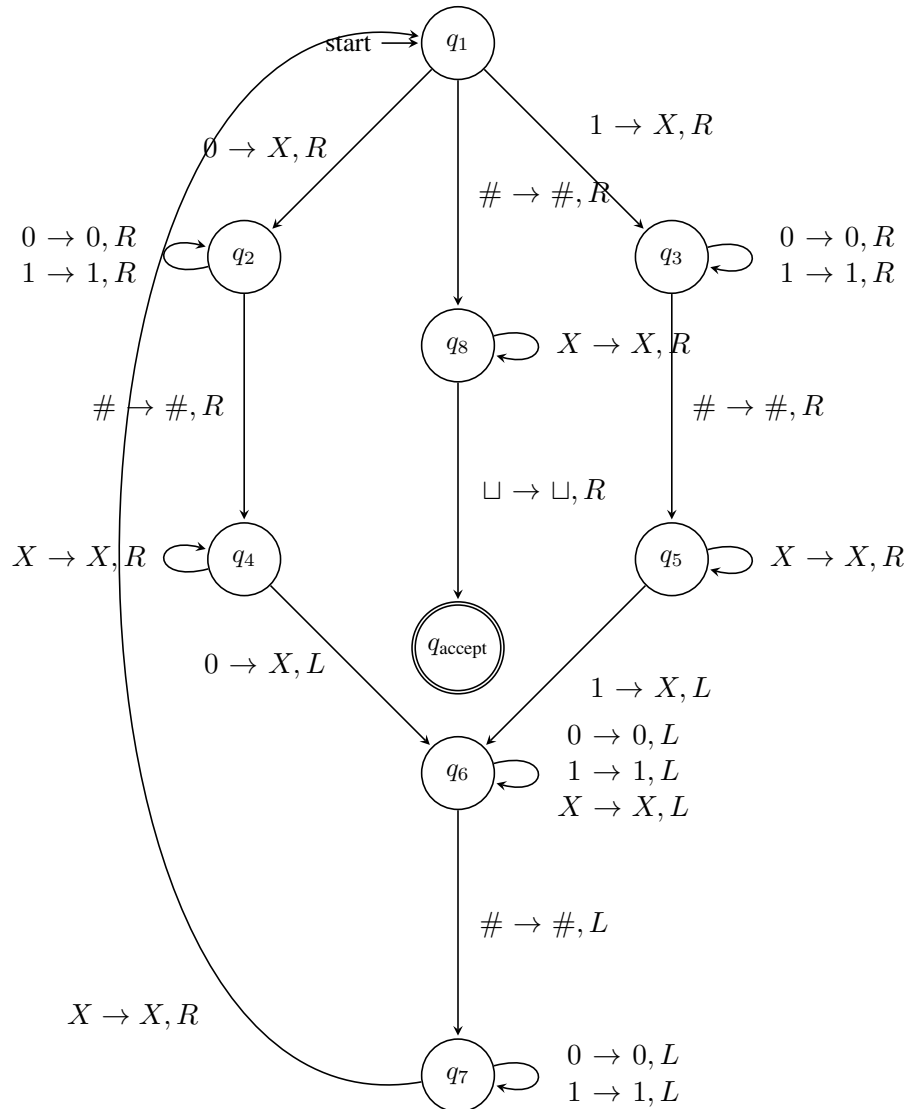
$q_1 0000$	$\sqcup q_2 000$	$\sqcup X q_3 00$	$\sqcup X 0 q_4 0$	$\sqcup X 0 X q_3 \sqcup$	$\sqcup X 0 q_5 X \sqcup$	$\sqcup X q_5 0 X \sqcup$
$\sqcup q_5 X 0 X \sqcup$	$q_5 \sqcup X 0 X \sqcup$	$\sqcup q_2 X 0 X \sqcup$	$\sqcup X q_2 0 X \sqcup$	$\sqcup X X q_3 X \sqcup$	$\sqcup X X X q_3 \sqcup$	$\sqcup X X q_5 X \sqcup$
$\sqcup X q_5 X X \sqcup$	$\sqcup q_5 X X X \sqcup$	$q_5 \sqcup X X X \sqcup$	$\sqcup q_2 X X X \sqcup$	$\sqcup X q_2 X X \sqcup$	$\sqcup X X q_2 X \sqcup$	$\sqcup X X X q_2 \sqcup$
						$\sqcup X X X \sqcup q_{\text{accept}}$

$$L(M_5) = \{w\#w \mid w \in \{0,1\}^*\}$$

Informal description:

1. Zig-zag across the tape to corresponding positions on either side of the $\#$ symbol to check whether these positions contain the same symbol. If they do not, or if no $\#$ is found, *reject*. Cross off symbols as they are checked to keep track of which symbols correspond.
2. When all symbols to the left of the $\#$ have been crossed off, check for any remaining symbols to the right of the $\#$. If any symbols remain, *reject*; otherwise, *accept*.

Formal description (transition diagram):



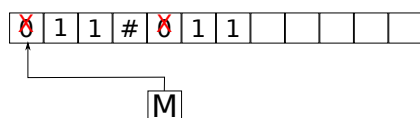
1.2 Convenient techniques in TM programming

We can assume the following when we talk about TMs:

1. The control unit (i.e., state machine) has a finite memory.
2. The tape has multiple tracks.
3. The TM can have another TM as a subroutine.

1.2.1 Control unit with a finite memory

Example A TM that accepts $\{w\#w \mid w \in \{0,1\}^*\}$



Here we remember we read 0, so we will process the next element of the second part if it is 0.

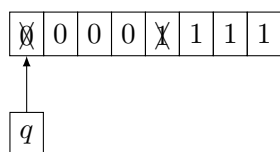
Note: This argument is valid only when the variable to be remembered has a finite domain. E.g. to accept $\{0^n 1^n \mid n \geq 0\}$, we cannot say “read the 0s and remember n ”.

1.2.2 Multi-track tape

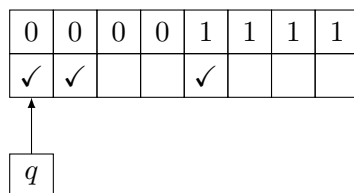
We can assume that the tape has multiple tracks.

E.g. a TM that accepts $\{0^n 1^n \mid n \geq 0\}$

Instead of this:



We can have:



If track-1 has alphabet Γ_1 , and track-2 has alphabet Γ_2 , a TM with a single-track tape with alphabet $\Gamma = \Gamma_1 \times \Gamma_2$ can do the same thing. E.g.

$$(0, \checkmark) = X$$

$$(1, \checkmark) = Y$$

$$(0, \sqcup) = 0$$

$$(1, \sqcup) = 1$$

1.2.3 Subroutines

A TM can use another TM as a subroutine. E.g., a TM that does multiplication can call another TM that does addition as a subroutine.

1.3 Variations of TMs

1. Multitape TM
2. Two-way infinite tape TM
3. Non-deterministic TM
4. ...

can all be emulated by a simple 1-tape, deterministic TM.

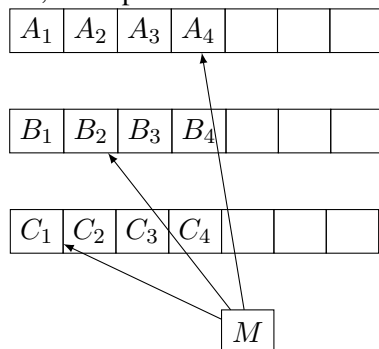
1.3.1 Multitape TM

- k tapes, k independent tape heads.
- At every transition:
 - a symbol is read from each tape.
 - state is changed.
 - on each tape, a new symbol is written, and the head is moved to the left or right.

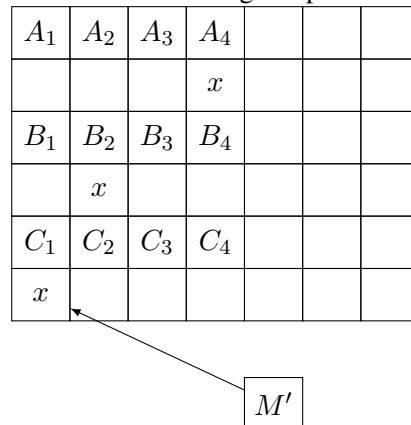
Theorem 1.1 *Every multitape TM has an equivalent single-tape TM.*

Proof by construction.

M , a k -tape TM:



M' : a $2k$ -track single-tape TM:

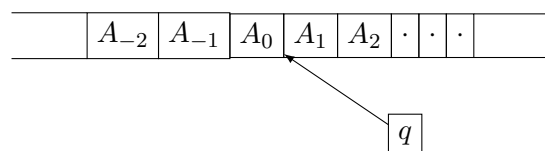


We write x at the position of our virtual tape. Then we can find where we left off on our tape easily.

M' emulates M as follows:

- The tape head makes k passes, once for each tape, locating the tape head emulator, and reading its symbol.
- Updates the state accordingly. Accepts if it is an accept state.
- Makes another k passes over the tape, locating each head emulator, updating its symbol, and moving the head emulator to the left or right. ■

1.3.2 Two-way infinite tape TM



It can easily be emulated by a 2-tape TM.

1.3.3 Non-deterministic TM

- Defined as expected:

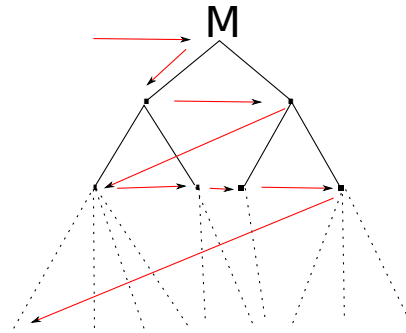
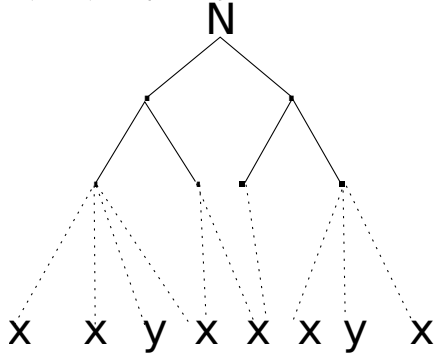
$$\delta : Q \times \Gamma \rightarrow 2^{Q \times \Gamma \times \{L, R\}}$$

- The computation of an NTM is a tree where different branches are different alternative paths.
- If some branch leads to an accept state, the input is accepted.

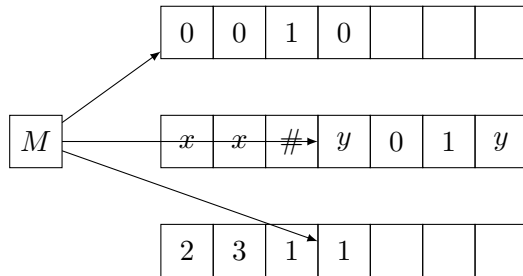
Theorem 1.2 *Every NTM has some equivalent DTM.*

Proof by construction.

$$\delta(q_0, a) = \{q_1, q_2\}$$



For an NTM N , we can construct a 3-tape TM M that emulates N and searches tree by **BFS** for an accepting configuration. Note that DFS may continue forever and never find an accepting state that is on a different branch.



Steps:

1. Copy the input to T_1 . Set T_2 and T_3 to be empty.
2. Copy T_1 to T_2 , set $T_3 = \epsilon$.
3. Evaluate T_2 , before each step, consult T_3 what to do (find the configurations). If $\square$ is reached in T_3 , abort, go to step 4.
4. Replace the string in T_3 with the next string (BFS steps), then go to step 2. ■

- T_1 : “input tape” to preserve the original input
- T_2 : “evaluation tape” emulates N on the input according to the branches specified by the address tape.
- T_3 : “address tape” to keep track of the BFS. 231 on the address tape means: go to the 2nd child of the root, then to its 3rd child, then to its 1st.

1.4 Other models equivalent to TMs

- Multi-stack PDA
- Counter machines with multiple counters